

**BADRAXH TARIALAN, Mongolia**

WMO: 442300

Lat: 49.6092N

Lon: 101.9847E

Elev: 1235

StdP: 87.34

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

**Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                   | (o)  | (p)   |
| 1             | -32.2      | -30.1 | -35.7                         | 0.1 | -32.0 | -33.8 | 0.2 | -29.6 | 7.7                   | -14.6 | 6.4 | -16.4 | 2.3                   | 20   | 0.654 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.9                   | 29.1            | 16.7 | 26.9 | 16.0 | 25.0 | 15.4 | 18.7                | 25.6 | 17.6 | 24.1 | 16.7 | 22.8 | 3.2                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.3                            | 13.5 | 22.0 | 15.3 | 12.6 | 20.8 | 14.3 | 11.8 | 19.9 | 58.6          | 25.7 | 54.8 | 24.1 | 51.7 | 23.0 | 25.0              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 8.9               | 7.3  | 6.2 | -33.4                      | 32.7 | 2.8                | 2.3 | -35.4                                              | 34.3 | -37.0      | 35.7 | -38.6      | 37.0 | -40.6      | 38.7 |    |    |
|                   |      |     | -33.5                      | 20.8 | 2.8                | 1.9 | -35.5                                              | 22.1 | -37.1      | 23.2 | -38.6      | 24.3 | -40.6      | 25.7 |    |    |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 0.7    | -20.6 | -15.6 | -5.4  | 4.1   | 10.4  | 16.2  | 18.2  | 16.0  | 10.0  | 1.4   | -9.7  | -17.8 |
|                                                                                  | DBStd       | 14.37  | 5.13  | 5.90  | 6.38  | 5.04  | 4.81  | 3.81  | 3.14  | 3.29  | 4.23  | 4.88  | 6.29  | 5.56  |
|                                                                                  | HDD10.0     | 4157   | 949   | 716   | 476   | 184   | 54    | 3     | 0     | 2     | 52    | 268   | 591   | 862   |
|                                                                                  | HDD18.3     | 6514   | 1207  | 949   | 734   | 427   | 247   | 84    | 40    | 86    | 252   | 526   | 841   | 1121  |
|                                                                                  | CDD10.0     | 757    | 0     | 0     | 0     | 7     | 68    | 188   | 254   | 188   | 51    | 1     | 0     | 0     |
|                                                                                  | CDD18.3     | 73     | 0     | 0     | 0     | 0     | 2     | 19    | 37    | 14    | 1     | 0     | 0     | 0     |
|                                                                                  | CDH23.3     | 816    | 0     | 0     | 0     | 1     | 69    | 257   | 347   | 133   | 9     | 0     | 0     | 0     |
|                                                                                  | CDH26.7     | 203    | 0     | 0     | 0     | 0     | 12    | 63    | 96    | 32    | 1     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.0    | 2.5   | 2.8   | 3.3   | 3.7   | 3.7   | 3.2   | 2.7   | 2.6   | 3.0   | 2.9   | 2.6   | 2.4   |
| Precipitation                                                                    | PrecAvg     | 302    | 2     | 1     | 3     | 8     | 20    | 55    | 102   | 74    | 24    | 7     | 3     | 3     |
|                                                                                  | PrecMax     | 438    | 15    | 8     | 10    | 28    | 60    | 144   | 170   | 142   | 53    | 21    | 12    | 11    |
|                                                                                  | PrecMin     | 207    | 0     | 0     | 0     | 0     | 0     | 3     | 22    | 18    | 2     | 0     | 0     | 0     |
|                                                                                  | PrecStd     | 64     | 3     | 2     | 3     | 6     | 14    | 33    | 34    | 34    | 14    | 6     | 3     | 3     |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | -6.0  | 1.0   | 13.8  | 21.9  | 28.4  | 31.3  | 32.2  | 30.2  | 24.5  | 16.9  | 7.2   | -3.7  |
|                                                                                  |             | MCWB   | -8.0  | -3.1  | 5.7   | 9.4   | 14.0  | 16.3  | 19.1  | 18.2  | 13.8  | 8.0   | 2.1   | -6.2  |
|                                                                                  | 2%          | DB     | -9.3  | -2.1  | 10.0  | 18.0  | 25.1  | 28.6  | 29.4  | 26.5  | 21.9  | 14.1  | 4.3   | -6.6  |
|                                                                                  |             | MCWB   | -10.9 | -5.3  | 2.9   | 7.7   | 12.2  | 15.5  | 18.1  | 17.1  | 12.9  | 6.5   | 0.2   | -8.5  |
|                                                                                  | 5%          | DB     | -11.4 | -4.6  | 7.2   | 15.5  | 22.6  | 26.3  | 27.0  | 24.3  | 19.6  | 11.6  | 1.7   | -8.8  |
|                                                                                  |             | MCWB   | -12.7 | -7.1  | 1.3   | 6.3   | 11.2  | 15.0  | 17.0  | 16.3  | 11.6  | 5.0   | -1.7  | -10.3 |
|                                                                                  | 10%         | DB     | -13.6 | -7.1  | 4.3   | 13.2  | 20.1  | 24.3  | 24.9  | 22.5  | 17.3  | 9.1   | -0.8  | -10.8 |
|                                                                                  |             | MCWB   | -14.6 | -9.1  | -0.4  | 5.1   | 10.0  | 14.4  | 16.7  | 15.6  | 10.5  | 3.6   | -3.4  | -12.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | -7.8  | -2.8  | 5.9   | 10.1  | 15.2  | 18.3  | 21.2  | 19.7  | 15.7  | 8.5   | 2.5   | -6.1  |
|                                                                                  |             | MCDB   | -6.0  | 0.5   | 13.4  | 19.6  | 25.8  | 27.1  | 30.3  | 26.0  | 21.9  | 16.1  | 6.5   | -3.8  |
|                                                                                  | 2%          | WB     | -10.8 | -5.2  | 3.3   | 8.3   | 13.3  | 16.9  | 19.3  | 18.1  | 13.8  | 6.9   | 0.3   | -8.4  |
|                                                                                  |             | MCDB   | -9.4  | -2.2  | 9.4   | 16.9  | 23.3  | 25.8  | 26.5  | 24.5  | 19.9  | 13.5  | 4.1   | -6.6  |
|                                                                                  | 5%          | WB     | -12.7 | -7.0  | 1.4   | 6.8   | 11.8  | 16.0  | 18.2  | 17.0  | 12.4  | 5.4   | -1.5  | -10.3 |
|                                                                                  |             | MCDB   | -11.5 | -4.7  | 6.8   | 14.7  | 21.6  | 23.7  | 24.7  | 23.0  | 18.3  | 11.3  | 1.5   | -8.9  |
|                                                                                  | 10%         | WB     | -14.6 | -9.0  | -0.4  | 5.4   | 10.3  | 15.0  | 17.3  | 16.0  | 11.0  | 3.9   | -3.4  | -12.0 |
|                                                                                  |             | MCDB   | -13.6 | -7.1  | 4.1   | 12.6  | 19.2  | 22.4  | 23.2  | 21.5  | 16.6  | 9.0   | -0.8  | -10.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.1   | 11.1  | 13.1  | 13.0  | 13.7  | 12.3  | 10.9  | 11.1  | 12.6  | 12.2  | 9.5   | 7.6   |
|                                                                                  |             | MCDBR  | 10.0  | 13.3  | 14.9  | 16.4  | 17.5  | 15.7  | 14.4  | 13.9  | 15.0  | 15.0  | 11.5  | 9.3   |
|                                                                                  |             | MCWBR  | 8.9   | 10.8  | 9.6   | 9.4   | 9.1   | 7.3   | 6.5   | 6.9   | 8.0   | 9.2   | 8.4   | 8.0   |
|                                                                                  | 5% WB       | MCDBR  | 9.9   | 13.0  | 14.4  | 15.3  | 16.2  | 13.2  | 12.3  | 12.1  | 13.5  | 14.3  | 11.2  | 9.2   |
|                                                                                  |             | MCWBR  | 8.8   | 10.7  | 9.6   | 9.4   | 9.3   | 6.9   | 6.5   | 6.7   | 7.6   | 9.1   | 8.4   | 7.9   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.238  | 0.262 | 0.307 | 0.347 | 0.375 | 0.375 | 0.365 | 0.343 | 0.301 | 0.282 | 0.260 | 0.241 |       |
|                                                                                  | taud        | 2.342  | 2.307 | 2.269 | 2.279 | 2.231 | 2.336 | 2.412 | 2.451 | 2.509 | 2.487 | 2.385 | 2.312 |       |
|                                                                                  | Ebn at Noon | 838    | 895   | 901   | 897   | 882   | 883   | 887   | 894   | 906   | 865   | 799   | 773   |       |
|                                                                                  | Edh at Noon | 64     | 86    | 109   | 121   | 133   | 121   | 110   | 101   | 85    | 72    | 62    | 58    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.64   | 2.76  | 4.23  | 5.39  | 6.32  | 6.39  | 6.08  | 5.25  | 4.38  | 2.93  | 1.75  | 1.27  |       |
|                                                                                  | RadStd      | 0.07   | 0.13  | 0.15  | 0.28  | 0.34  | 0.40  | 0.31  | 0.27  | 0.21  | 0.12  | 0.08  | 0.07  |       |

**Historical Trends**

|                        |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                    | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| Station Only           | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     |         |         |
| Regional (5 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | -0.70 | N/A   | N/A         | N/A     |         |         |

Nomenclature: See separate page